

Lymphedema

The Big Picture

Lymphedema = **chronic** swelling from too much lymph fluid — either **impaired outflow (obstruction)** or **overproduction exceeding transport capacity**. Protein-rich fluid pools → fibrosis → skin changes → eventually **elephantiasis**. No cure; management is lifelong. Don't confuse with **lipedema** (fat) or **venous edema**.

Lymph Basics

- Lymph = WBCs, triglycerides, cell debris, water, **protein**.
- Drainage chain: capillaries → precollectors → collectors → trunk → nodes.

Classification — Primary vs Secondary

Primary (1 in 6000–10,000 births; **females 2–20×**, peak 12–16 yrs). Sub-classified by:

- **Time of onset:**
- **Congenital** — at birth/first year; sporadic or AD (Milroy); linked to **Turner, Klinefelter, Trisomy 21**; bilateral whole-leg.
- **Praecox (Meige's)** — at puberty/3rd decade; **most common (up to 94%)**; F:M **10:1**; unilateral foot+calf; oestrogen role.
- **Tarda** — after **35 yrs**; ~10% of primary.
- **Morphology:** aplasia (none), hypoplasia (few — commonest), numerical hyperplasia, hyperplasia (megalymphatics — worst prognosis).
- **Anatomy:** distal obstruction (■; bilateral mild, female, good prognosis) vs proximal obstruction (½; unilateral severe, relentless, may need surgery).

Secondary (acquired) — most common overall. Causes:

- **Cancer** — breast cancer treatment (**axillary dissection + radiotherapy**) → upper-limb; gynae/urologic → lower-limb/pelvis; melanoma after radical dissection.
- **Filariasis** — *Wuchereria bancrofti* (90%), Brugia; commonest cause in tropics; **90 million infected**; adult worms → fibrosis; **eosinophilia**, microfilariae in **nocturnal blood**; → gross **elephantiasis**.
- **Trauma** — burns, circumferential wounds, infection, pregnancy.

Clinical Staging

- **Latent** — fibrosis around lymphatics, no visible edema.
- **Grade 1** — **pits**, reduced by elevation, no fibrosis.
- **Grade 2** — **non-pitting**, not reduced by elevation, marked fibrosis.
- **Grade 3** — irreversible edema + sclerosis = **elephantiasis**.

Clinical Signs

- Soft pitting edema early (protein-rich).
- **Stemmer's sign positive** — can't tent the skin at base of 2nd toe (hallmark).
- **Square toes, buffalo hump** (foot dorsum), woody texture over time.
- Skin: pinkish-red early → thick, hyperkeratosis, lichenification, **peau d'orange**; lymphorrhea.
- **Yellow nail syndrome** = primary lymphedema + yellow nails + pleural effusion.
- Pain usually just aching/heaviness — intense pain → suspect infection.

Complications

- **Recurrent infection / cellulitis** — most troublesome; worsens edema.
- Malnutrition + immunodeficiency (protein-losing enteropathy, chylous ascites/thorax).
- **Lymphangiosarcoma** — rare malignancy, may need amputation.

Diagnosis

- Exam (Stemmer, skin), **isotopic lymphoscintigraphy** (key test), near-infrared fluorescence, MRI/CT, US.
- **DDx**: lipedema, venous insufficiency, systemic bilateral pitting (cardiac/renal/liver/hypoproteinemia/myxedema), Klippel-Trenaunay.

Management (no cure)

- **Conservative first line — Complex Decongestive Therapy**: manual lymphatic drainage + compression bandaging + compression garments + skin care + exercise; lifelong.
- **Operative** (last resort) — volume reduction/liposuction; reconstruction (**lymphovenous anastomosis**, lymphatic grafting).

Walk-Away Points

- **Praecox (Meige's) = commonest primary** (94%), puberty, F:M 10:1.
- **Secondary is commonest overall**; breast-cancer axillary surgery (arm) + **filariasis** (tropics) are top causes.
- **Stemmer's sign** = the clinical hallmark.
- Staging: pits (G1) → non-pitting fibrosis (G2) → elephantiasis (G3).
- Lifelong **complex decongestive therapy**; surgery last.

Lymphedema — Revision Table	
Basics	Detail
Definition	Chronic swelling: impaired outflow OR overproduction
Lymph content	WBC, triglycerides, debris, water, protein
Primary epidemiology	1:6000–10,000; females 2–20× , peak 12–16
Primary — onset	Features
Congenital (Milroy)	Birth/1st yr; Turner, Klinefelter, T21; bilateral
Praecox (Meige's)	Puberty; commonest (94%) ; F:M 10:1 ; unilateral foot/calf
Tarda	After 35 yrs ; ~10%
Primary — anatomy	Features
Distal obstruction	■, bilateral, mild, female, good prognosis
Proximal obstruction	½; unilateral, severe, relentless → surgery
Secondary (commonest overall)	Detail
Cancer	Breast (axillary dissection + RT) → arm; gynae/uro → leg
Filariasis	*W. bancrofti* (90%); tropics; eosinophilia; nocturnal microfilariae → elephantiasis
Trauma	Burns, wounds, pregnancy
Stage	Finding
Latent	Fibrosis, no visible edema
Grade 1	Pits , reduced by elevation
Grade 2	Non-pitting , fibrosis
Grade 3	Irreversible = elephantiasis
Signs	Detail
Stemmer's sign	Can't tent skin at base of 2nd toe (hallmark)
Square toes / buffalo hump	Foot involvement
Skin	Peau d'orange, hyperkeratosis, lichenification
Yellow nail syndrome	Primary LE + yellow nails + pleural effusion
Complications	Infection/cellulitis (commonest), immunodeficiency, lymphangiosarcoma
Management	Detail
First line	Complex decongestive therapy (MLD + compression + skin care)
Investigation	Lymphoscintigraphy
Operative (last)	Liposuction, lymphovenous anastomosis
DDx	Lipedema, venous, systemic pitting edema, Klippel-Trenaunay

Bold = high-yield.